



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)
BSc CHEMISTRY**

Programme	B.Sc Chemistry				
Course Title	POLYMERS IN INDUSTRY				
Type of Course	VOCATIONAL MINOR				
Semester	VIII				
Academic Level	300-399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4			60
Pre-requisites					
Course Summary	“Polymers in Industry” offers an in-depth exploration of the wide-ranging applications of polymers across various industrial sectors				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Students should gain insight into the diverse range of applications and uses of natural fibers across industries such as textiles, apparel, home furnishings, agriculture, packaging, construction, and craft industries	R, U, Ap	C	Class test /Assignment / Quiz
CO2	Students will conduct a comparative environmental analysis between natural and synthetic fibers and they will evaluate the life cycle environmental impacts of both fiber types	Ap,An, E	C,M	Class test /Assignment / Seminar
CO3	Students will become familiar with basic terminology related to adhesives, including terms such as viscosity, cure time, tack, cohesive strength, adhesive failure, and cohesive failure	U	C	Class test /Assignment / Quiz
CO4	Students will explore theories of adhesion, including mechanical interlocking, electrostatic attraction, diffusion, and chemical bonding theories. They will understand how these theories contribute to	U, Ap	C	Class test /Assignment / seminar

	the understanding of adhesive bonding at the molecular level.			
CO5	Students will develop a comprehensive understanding of pigments and paints, including their composition, properties, and applications in various industries such as coatings, plastics, printing, and cosmetics	U, Ap	C	Class test /Assignment / seminar
CO6	Develop a comprehensive understanding of food packaging materials, including conventional materials (e.g., plastics, metals, glass) and emerging edible and biobased materials. They will learn about the properties, functionalities, and applications of different packaging materials in the food industry	R,U	C	Class test /Assignment / Viva
CO7	Develop a thorough understanding of polymers used in biomedical applications, including their chemical structures, properties, and biocompatibility and students will learn about polymer types such as biodegradable polymers, hydrogels, elastomers, and nanocomposites	R,U,Ap	C	Class test /Assignment / seminar
CO8	Students will develop a thorough understanding of the properties of polymers relevant to aerospace applications, including mechanical properties (e.g., strength, stiffness, toughness), thermal properties (e.g., heat resistance, thermal expansion), chemical resistance, and durability in harsh environments (e.g., UV radiation, moisture, space vacuum)	R,U,Ap	C,M	Class test /Assignment / Quiz
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Mark
I	Polymers in fiber industry		12	24
	1	Harvesting, extraction, processing, and characteristics of natural fibers (cotton, wool, silk, jute, flax).	2	
	2	Application and uses of natural fibers in textiles. Regenerated cellulose fibers-viscose,tencel, cellulose acetate and triacetate (Mention only)	2	
	3	Polyester, nylon, acrylic, polypropylene, polyethylene, acetate, Lycra and their manufacturing processes. Properties and advantages of synthetic 2 fibers.	3	

	4	Comparison with natural fibers in terms of properties and applications. Environmental considerations in synthetic fiber production.	2	
	5	Other fiber forming materials -glass, ceramic, carbon and metal. Innovations in fiber technology. Sustainable practices in textile fibers.	3	
		Polymers in adhesives coating	15	30
	6	Adhesives-adhesive bonding, advantages-adhesive classification basic terminology, theories of adhesion-wettability	2	
	7	Performance of adhesives - shear, peel and cleavage properties, factors affecting adhesive performance	2	
	8	Design of adhesive joints, selection of adhesives. Structural adhesive, types - epoxy, urethane, acrylic, phenolic and high temperature and PVC plastisol types	2	
	9	Advantages and disadvantages, anaerobic adhesives, cyanoacrylates, hot melt adhesive, pressure sensitive adhesives, silicone adhesives, water based adhesives, inorganic adhesives	2	
	10	Pigments and paints, inorganic pigments & organic pigments, extenders paint preparation factors affecting dispersion, preparation of pigment dispersion	2	
	11	Surface preparation and paint application techniques. Paint properties and their evaluation, mechanism of film formation factors affecting coating properties, methods used for film preparation	3	
	12	Carrier properties, optical properties, ageing properties, rheological properties and adhesion properties of coatings.	2	
		PACKAGING APPLICATIONS	10	22
	13	Edible and biobased food packaging materials, Edible film and coating,	2	
	14	Polysaccharide based coatings, Lipid based coatings, Protein based coating, First, Second and Third biobased packaging materials.	2	
	15	Permeability of thermoplastic polymers, Multilayer films, Processing, Deteriorative reaction in foods, Enzyme reactions, Chemical reactions,	2	
	16	Physical change, Biological change, shelf life of foods, Factors controlling 2 shelf life.	2	
	17	Packaging of dairy products, Packaging of cereals, snack foods and confectionary, Packaging of beverages, Comparison of polymer packaging with paper, metal and glass materials	2	
IV		POLYMERS IN BIOMEDICAL APPLICATION	11	22

	18	Definition of biomedical Polymers and its classification, Criteria for the Selection of Biomedical Polymers	2	
	19	Properties of biomedical Polymers, : Polymers for biomedical applications– Polymers in dentistry	2	
	20	polymers in Tissue adhesives , Dialysis Membrane	2	
	21	Polymers in Blood oxygenators, Bone cement, Prostheses	2	
	22	polymers in Biodegradable sutures, Control drug delivery systems	3	
V		POLYMERS IN AEROSPACE APPLICATION	12	
	24	Requirement of polymer characteristics for in space usage, polymers in aerospace applications: thermal blanket, helmet,	4	
	25	polymers in aerospace applications: Adhesive, Space Suits	3	
	26	polymers in aerospace applications: Eelectronic applications, structural applications Important plastics used in Aerospace industry (Thermosetting polyimide, Polyetheretherketone,Polyamide-imide,Polycholotriflouroethylene,PTFE	2 3	

References

1. Gowariker, V.R., Viswanathan, N.V., Sreedhar, J., *Polymer Science*, Wiley, New Delhi, India, 1990.
2. Billmeyer, F.W., *Textbook of polymer science*, Wiley, New Delhi, India, 2007.
3. Deopura, B.L., Alagirusamy, R., Joshi, M., Gupta, B., *Polyesters and polyamides*, Woodhead Publishing Limited, Cambridge, UK, 2008.
4. Ebewele, R.O., *Polymer science and technology*, CRC Press, New York, USA, 2000.
5. Cook, J.G., *Handbook of textile fibers natural fibers*, p. 240, Woodhead Publishing, Cambridge, UK, 1984.
6. W. C wake, *Adhesives and the Formulation of Adhesives*, Applied science Publishers, London,1976.
7. Yosomiya R, Morimoto K, Nakajima A, Ikada Y, Suzuki T. Adhesionand bonding in composites. 1st ed. New York: Marcel Dekker Inc.;1989.
8. R. Sinha,outlines Polymer Technology, Prentice Hall India,2000.
9. P. M Ajayan, L.S Schadler and P. V Braun, nanocomposite Science and Technology, Wiley-VCH GmbH Co.
10. Ito K. Structural adhesives for aerospace applications. Int Poly Sci Technol, vol 14, no 8, 1987, pp 37-43

11. Thomas D K. Uses of rubber and composites in aerospace. Plast Rubber Int, vol 8, no 2, April 1983, pp 53-57

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	-	1	1	2	3	3		2	1	2	2	3
CO 2	2		2	2	1	1	3	1	2	1	2	3	2
CO 3	3	-	2	-	-	-	3					2	2
CO 4	3	-	2	1	-	-	3					1	1
CO 5	3		1	2	-	-	3		1		1	2	2
CO 6	3	-	1	2	1	1	3		1		2	2	2
CO 7	3			2	2	3	3		2		2	3	3
CO 8	3			3	2	3	3		2		2	3	3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6	✓	✓		✓
CO7	✓	✓		✓
CO8	✓	✓		

SKILL ENHANCEMENT COURSES